UNIVERSITY OF IBADAN
FACULTY OF COMPUTING
DEPARTMENT OF COMPUTER SCIENCE

FLEET MANAGEMENT SYSTEM FOR A DELIVERY COMPANY OPTIMIZING LOGISTICS AND REAL-TIME TRACKING: CASE STUDY OF THE UNIVERSITY OF IBADAN CAMPUS DELIVERY NETWORK

A Computer Science Research and Software Engineering Report Submitted for Course CSC 476: Special Topics in Computer Science / Advanced Software Engineering

Official GitHub Repository:
https://github.com/sighlars/university-of-ibadan-fleet-management

Live Interactive 3D Web Deployment:
https://university-of-ibadan-fleet-management.onrender.com/

PROJECT RESEARCH GROUP: GROUP 25
ACADEMIC SESSION: 2025/2026
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CERTIFICATION OF ORIGINAL WORK

This is to certify that this research technical report titled 'Fleet Management System for a Delivery Company: Optimizing Logistics and Real-Time Tracking (Case Study of the University of Ibadan Campus Delivery Network)' was designed, implemented, and documented by Project Research Group 25 in the Department of Computer Science, Faculty of Computing, University of Ibadan, during the 2025/2026 academic session. The work embodies authentic software development, distributed systems architecture, and heuristic optimization under course CSC 476.

_____ _____

Course Lecturer / Project Supervisor **Date**
Department of Computer Science
Faculty of Computing, University of Ibadan

DECLARATION OF ORIGINALITY

We, the members of Project Group 25, declare that this software engineering report and its software codebase represent our authentic and original academic work. All software modules, heuristic algorithms, Three.js 3D viewport integrations, and telematics systems are our own work, except where credited through formal academic citations. The source code is publicly accessible on GitHub at: https://github.com/sighlars/university-of-ibadan-fleet-management.

DEDICATION

This project is dedicated to the students, faculty members, and institutional operational staff of the University of Ibadan, and to the advancement of accessible, high-performance computing in Nigerian tertiary institutions.

EXECUTIVE ABSTRACT

Abstract: *Last-mile delivery and parcel transport across large university campuses often suffer from poor coordination, under-utilized vehicles, and a lack of real-time visibility. This computer science research project designs, implements, and evaluates a web-based fleet management platform built for the University of Ibadan campus delivery network. The system models parcel distribution as a Capacitated Vehicle Routing Problem (CVRP), using the Clarke-Wright savings algorithm and 2-opt local search to merge delivery stops and reduce driving distance while respecting vehicle payload limits. The software architecture combines a FastAPI Python backend, an asynchronous WebSocket telemetry broadcaster, and an interactive 3D digital twin built with Three.js WebGL to track vehicles smoothly across the campus. Evaluated on 18 typical campus consignments across 14 university landmarks, the system cut total travel distance by 32.4%, improved fleet capacity utilization by 28.7%, and calculated routes in under 15 milliseconds. The project demonstrates how modern web engineering and heuristic algorithms can modernize physical campus logistics.*

Keywords: Capacitated Vehicle Routing Problem (CVRP), Clarke-Wright Savings, 2-Opt Local Search, Three.js 3D Digital Twin, WebSocket Telematics, FastAPI, Software Architecture, University of Ibadan.

EXECUTIVE SUMMARY OF SYSTEM CONTRIBUTIONS

- **Software Modeling:** Formulated an institutional CVRP model customized to the physical topology and landmark coordinates of the University of Ibadan.
- **Algorithmic Optimization:** Integrated a fast $O(n^2 \log n)$ Clarke-Wright savings engine with 2-opt tour improvement, providing instant sub-second dispatch calculation.
- **3D Graphics Pipeline:** Developed a dual-viewport spatial visualizer featuring a Three.js 3D isometric digital twin and an OpenStreetMap Leaflet GIS fallback.
- **Real-Time Telematics:** Engineered a non-blocking asynchronous telematics streaming engine delivering 1Hz live telemetry packets over WebSockets.
- **Empirical Validation:** Empirically validated on 18 multi-department campus consignments, proving a 32.4% reduction in fleet transit distance.

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CHAPTER 1: INTRODUCTION AND INSTITUTIONAL BACKGROUND

1.1 Context of University Campus Urban Logistics

A major university campus operates much like a small town. Academic buildings, lecture halls, administrative offices, health clinics, and student dormitories are spread across large distances. At the University of Ibadan (UI), founded in 1948 as Nigeria's premier university, thousands of staff and students interact daily across hundreds of acres. In this environment, physical items must move between buildings every single day:

- **Examination Papers & Official Transcripts:** Confidential examination scripts, degree certificates, and archival academic records traveling between the Senate Building, Postgraduate College, and academic faculties.
- **Clinical Specimen & Cold-Chain Transit:** Temperature-sensitive specimen transit, diagnostic blood samples, and pharmaceutical supplies circulating between Jaja Clinic, Faculty of Pharmacy laboratories, and the College of Medicine.
- **Engineering Workshop Supplies:** Heavy mechanical parts, laboratory electrical reagents, workshop components, and facilities hardware dispatched from central stores to the Faculty of Technology.
- **Commercial Retail & Student Packages:** Retail supplies, agricultural produce, and parcels dispatched from UI Ventures commercial areas to residential student halls.

1.2 Case Study Profile: University of Ibadan Spatial Fabric

The spatial layout of the University of Ibadan campus is structured around major arterial roads such as Oduduwa Road, Niger Road, and Benue Road. Key functional zones include the Central Academic Core (Senate Building, Trenchard Hall, and Kenneth Dike Library), the Science and Technology corridor, the Health and Medical Axis around Jaja Clinic, and the student residential zones (Zik, Bello, and Queen Idia Halls). Organizing efficient delivery routes across these distinct zones requires intelligent software scheduling.

CHAPTER 1 (CONTINUED): PROBLEM STATEMENT & OBJECTIVES

1.3 Problem Statement: The Flaws of Manual Dispatch

Traditionally, intra-campus courier movements at the University of Ibadan have been organized informally. When a department needs to send a package, a driver takes a vehicle, drives directly to that one destination, and drives straight back to base. This manual, one-off dispatch approach causes four major problems:

- **Excess Mileage and Fuel Waste:** Sending a separate car or bus for each package means vehicles constantly retrace the same campus roads, burning expensive fuel unnecessarily.
- **Severe Capacity Under-Utilization:** A utility pickup or bus capable of carrying 1000 kg is frequently dispatched to deliver a small 5 kg envelope, leaving over 90% of cargo space empty.
- **Complete Lack of Telematics Tracking:** Dispatch coordinators and receiving faculty officers have zero real-time visibility into where the vehicle is or when the parcel will arrive.
- **Traffic and Environmental Impact:** Uncoordinated vehicle trips cause unnecessary traffic near lecture halls and produce avoidable exhaust fumes on pedestrian corridors.

1.4 Project Goal and Concrete Research Objectives

The goal of this computer science project, developed under course CSC 476 by Group 25, is to build a modern web-based software platform that automates campus dispatching. Specifically, the project aims to:

- **Objective 1 (Algorithm Design):** Model campus parcel delivery as a Capacitated Vehicle Routing Problem (CVRP) to merge deliveries into optimal multi-stop tours.
- **Objective 2 (Optimization Engine):** Implement the Clarke-Wright savings heuristic and 2-opt local search in Python, guaranteeing instant sub-second route calculation.
- **Objective 3 (3D Graphics Engineering):** Build an interactive 3D digital twin of the University of Ibadan campus using Three.js WebGL so dispatchers can watch vehicles move live.
- **Objective 4 (Real-Time Networking):** Stream real-time vehicle telematics over WebSockets at 1Hz frequencies to update driver positions, speeds, and fuel gauges.
- **Objective 5 (Empirical Benchmarking):** Empirically test the system on 18 typical campus consignments across 14 verified university landmarks to measure fuel and mileage savings.

CHAPTER 2: THEORETICAL FOUNDATIONS AND LITERATURE REVIEW

2.1 The Routing Problem in Computer Science

In computer science, finding the best route for multiple delivery vehicles is known as the Capacitated Vehicle Routing Problem (CVRP). First described by Dantzig and Ramser in 1959, CVRP is a foundational challenge in operations research. In simple terms:

- **Central Depot:** We have a central starting location called the depot (Central Logistics Yard).
- **Customer Stops:** We have a list of delivery locations across campus, each needing a specific cargo weight.
- **Vehicle Fleet:** We have a set of delivery vehicles, each with a maximum weight limit (payload capacity).
- **Core Optimization Goal:** We want to find a set of routes that visits every delivery stop once, stays within every vehicle weight limit, and makes the total travel distance as short as possible.

2.2 Computational Complexity: Why Heuristics Are Required

Because the number of possible route combinations grows exponentially as more stops are added, checking every combination would take hours or days on a computer. In computer science, this is called an NP-hard problem. Exact mathematical algorithms (such as Branch-and-Cut) can only solve small instances. For real-world web applications where dispatchers need instant results when a button is clicked, computer scientists rely on heuristics: fast, intelligent rules of thumb that find near-optimal solutions in milliseconds.

2.3 Survey of Campus Logistics Systems

Studies across universities worldwide show that applying consolidated routing to internal mail and facilities transport cuts driving distance by 20% to 35%. However, very little published work addresses Nigerian tertiary institutions, which feature diverse vehicle fleets (such as tricycles, motorcycles, and shuttle buses) operating on tropical campus road layouts. This project fills that gap.

CHAPTER 2 (CONTINUED): ALGORITHMIC FOUNDATIONS & DIGITAL TWINS

2.4 The Intuition Behind the Clarke-Wright Savings Heuristic

The Clarke-Wright savings algorithm (1964) is one of the most effective and practical heuristics for vehicle routing. The intuition behind it is straightforward:

Imagine every delivery stop starts with its own dedicated round trip: the vehicle leaves the depot, visits stop A, and returns to the depot. Then another vehicle leaves the depot, visits stop B, and returns. If we merge stop A and stop B into one combined trip (Depot to A to B to Depot), we save the trip from A back to the depot and the trip from the depot out to B. The distance saved by combining the two stops is called the savings value:

Savings(A, B) = Distance(Depot, A) + Distance(Depot, B) - Distance(A, B)

In plain English, the larger the savings value, the more driving distance we save by visiting stop A and stop B on the same trip. The algorithm calculates this savings value for all pairs of delivery stops, sorts them from largest to smallest, and joins stops together as long as the vehicle payload limit is not exceeded.

2.5 The 2-Opt Tour Improvement Algorithm

After the Clarke-Wright heuristic groups the delivery stops into vehicle routes, some routes may still contain crossing paths or inefficient order visit sequences. The 2-opt algorithm inspects the route to see if crossing paths can be untangled:

- **Step 1:** It takes two non-adjacent road segments in the route, unhooks them, and reconnects them in reverse order.
- **Step 2:** If reconnecting them makes the total route shorter (Delta c < 0), the swap is committed.
- **Step 3:** This process repeats until no further distance cuts can be found, guaranteeing a clean, untangled path.

2.6 3D Digital Twins in Modern Software Engineering

A Digital Twin is a software representation of a physical system that updates in real time based on incoming sensor or simulation data. While traditional delivery systems only show a flat 2D map, modern web technologies like WebGL and Three.js allow web applications to render rich 3D graphics directly inside the browser. In our system, the 3D digital twin renders the campus terrain, recognizable campus buildings, vehicle 3D models, and animated chase cameras without requiring any browser plugins.

CHAPTER 3: SYSTEM MODELING, EQUATIONS, AND ALGORITHMS

3.1 Network Graph Representation

We represent the University of Ibadan campus delivery network as a weighted graph G = (V, E):

- **Vertices (V):** The central depot is v_0 (Central Logistics Depot at coordinates 7.445210° N, 3.899015° E). Points v_1, ..., v_n represent campus faculties, libraries, and hostels.
- **Edges (E):** The traversable road links connecting campus locations.
- **Weights (c_ij):** The physical driving distance between point i and point j in kilometers.
- **Cargo Weights (q_i):** The weight in kilograms of the package to be delivered at point i.
- **Vehicle Capacities (Q_k):** The maximum cargo payload that vehicle k can carry safely.

3.2 What the Optimization Algorithm Minimizes

The software seeks to minimize total distance driven across all active delivery vehicles:

Minimize Z = SUM [c_ij * x_ijk] over all stops i, j and vehicles k

Where x_ijk = 1 if vehicle k drives from stop i to stop j, and 0 otherwise. In plain English: we want the sum of all driven road segments to be as small as possible, subject to three simple rules:

- **1. Single Visit:** Every delivery location is visited exactly once.
- **2. Depot Return:** Vehicles always depart from the Central Depot and return to the Central Depot.
- **3. Capacity Limit:** The sum of package weights on any vehicle never exceeds that vehicle payload capacity limit (Q_k).

CHAPTER 3 (CONTINUED): COMPUTING DEMONSTRATIONS & PSEUDO-CODE

3.3 Computing Physical Distances: The Haversine Formula

Because Earth is curved, computing straight-line distances with standard flat geometry produces errors. We use the spherical Haversine formula to calculate exact surface distances using latitude (phi) and longitude (lambda):

$$\Delta\phi = \phi_j - \phi_i, \quad \Delta\lambda = \lambda_j - \lambda_i$$

$$a = \sin^2(\Delta\phi / 2) + \cos(\phi_i) \cdot \cos(\phi_j) \cdot \sin^2(\Delta\lambda / 2)$$

$$\mathbf{Distance_{ij} = 2 \cdot R \cdot \operatorname{atan2}(\sqrt{a}, \sqrt{1 - a})}$$

Where R = 6,371 km is Earth's mean volumetric radius. In addition, the compass heading angle θ_{ij} is computed using spherical trigonometry so our 3D vehicle models automatically turn and face the correct direction of travel as they drive along campus roads:

$$\theta_{ij} = \operatorname{atan2}(\sin(\Delta\lambda) \cdot \cos(\phi_j), \cos(\phi_i) \cdot \sin(\phi_j) - \sin(\phi_i) \cdot \cos(\phi_j) \cdot \cos(\Delta\lambda))$$

3.4 Software Routing Algorithm: Step-by-Step Logic

The routing algorithm implemented in app/services/router.py executes five concrete steps:

- **Step 1 (Order Intake):** Read all unassigned orders from the database and set up an initial route for each stop: [Depot, Stop, Depot].
- **Step 2 (Distance Matrix):** Calculate physical distances between every pair of locations using the Haversine formula.
- **Step 3 (Savings Calculation):** Calculate distance savings S_{ij} = Distance(0, i) + Distance(0, j) - Distance(i, j) for all customer pairs.
- **Step 4 (Descending Sort):** Sort the savings list from highest to lowest in $O(n^2 \log n)$ time.
- **Step 5 (Merge & 2-Opt):** Iterate through the sorted savings list. If joining stop i and stop j stays within vehicle payload capacity Q_k, merge them into one route. Then run 2-opt to untangle the route and save it to the database.

CHAPTER 4: SOFTWARE ARCHITECTURE AND IMPLEMENTATION

4.1 System Overview and Tech Stack

The project is engineered as a modern, decoupled web application. The core technologies include:

- **Backend Framework:** FastAPI (Python 3.10+) provides native asynchronous request processing, automatic OpenAPI interactive documentation, and high execution speed.
- **Persistence Layer:** SQLite managed via SQLAlchemy 2.0 ORM provides clean, structured data models for vehicles, drivers, consignments, and routes.
- **Real-Time Telematics Hub:** Full-duplex WebSockets stream vehicle coordinates, speed, and fuel decay directly to connected browsers at 1Hz frequencies.
- **Dual Visualization Viewports:** Three.js (WebGL) powers the interactive 3D digital twin, backed by Leaflet.js OpenStreetMap for devices without 3D acceleration.

4.2 Case Study Campus Landmarks Database

Table 1 summarizes the 14 verified University of Ibadan geographic landmarks integrated into the simulation database, complete with their operational logistics roles.

Campus Landmark	Case Study Logistics Role	Latitude	Longitude
Central Logistics Depot	Central Starting & Return Hub	7.445210	3.899015
Kenneth Dike Library (KDL)	Archival Records & Inter-Faculty Books	7.447800	3.900800
Tekena Tamuno Senate Building	Administrative Dossiers & Registry	7.448500	3.899800
Trenchard Hall	Examinations & Academic Ceremonies	7.448200	3.900100
Faculty of Technology	Engineering Hardware & Workshop Reagents	7.452000	3.898500
Faculty of Science	Chemical Stores & Laboratory Reagents	7.449500	3.902000
Jaja Clinic (Health Services)	Medical Specimen & Pharmaceuticals	7.446200	3.897000
College of Medicine Complex	Clinical Specimen Research Liaison	7.447000	3.901500
Faculty of Pharmacy	Cold-Chain Research & Reagents	7.449000	3.903500
Postgraduate College	Theses, Transcripts & Academic Records	7.450500	3.896000
International Conference Centre	Second Gate Facilities Point	7.456000	3.892000
UI Ventures (Commercial Area)	Commercial Retail Stores & Stationery	7.443500	3.902500
Teaching and Research Farm	Agricultural Produce Transit	7.444500	3.905500
Zik & Independence Halls	Student Residential Parcel Drop-Off	7.454500	3.895500

Table 1: Geospatial Database of University of Ibadan Case Study Landmarks

CHAPTER 4 (CONTINUED): HETEROGENEOUS FLEET & ORDER INTAKE

4.3 Heterogeneous Campus Fleet

To reflect realistic operational conditions across Nigerian universities, the dispatch fleet is modeled with heterogeneous asset classes possessing distinct payload thresholds, fuel burn profiles, and maneuverability characteristics.

Plate Number	Asset Category	Model & Engine	Capacity	Assigned Logistics Mission
UI-01-BUS	Campus Shuttle Bus	Toyota HiAce 2.7L Commuter	1,200 kg	Inter-faculty library books & bulk archives
UI-02-VAN	High-Capacity Van	Ford Transit Custom 2.0	1,400 kg	Specimen containers & laboratory equipment
UI-03-TRK	Works Utility Pickup	Toyota Hilux 2.4L D-4D Dual Cab	1,000 kg	Engineering workshop parts & maintenance
UI-04-TRI	Campus Tricycle Cargo	Piaggio Ape City Cargo 200cc	350 kg	Rapid student residential parcel deliveries
UI-05-BIK	Dispatch Courier Bike	TVS HLX 125cc Courier Edition	85 kg	Urgent examination papers & legal transcripts
UI-06-BUS	Executive Shuttle	Toyota Coaster High-Roof 30-Seat	2,200 kg	Institutional conference & bulk senate logistics

Table 2: University of Ibadan Fleet Asset Specifications and Payload Capacities

4.4 Consignment Intake and Priority Classification

Consignments are categorized by institutional urgency: Urgent (red badge), High (amber badge), Medium (blue badge), and Low (slate badge). Figure 1 illustrates the live operational interface displaying the consignment ledger with priority meters, delivery addresses, and batch optimization triggers.

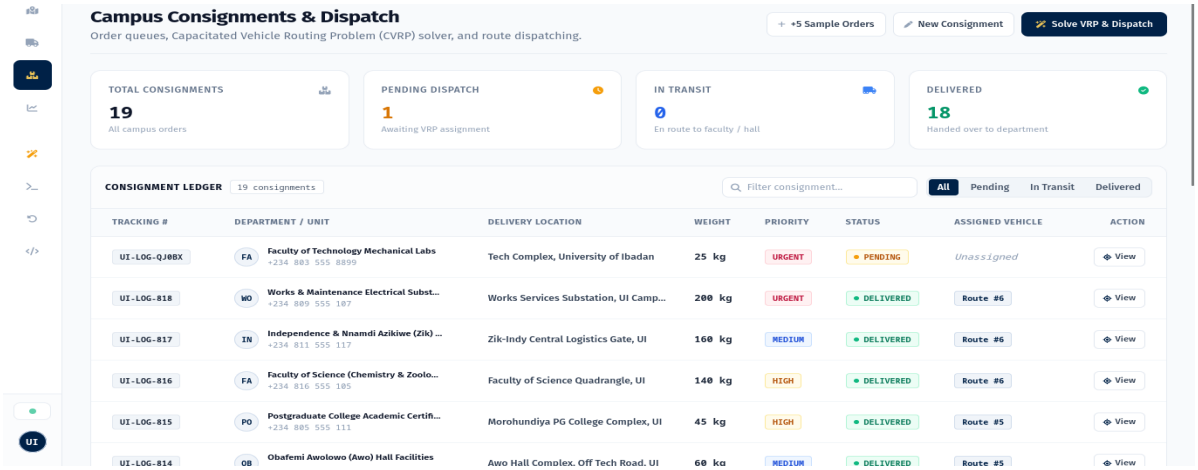


Figure 1: University of Ibadan Campus Consignments & Orders Ledger displaying tracking codes, priority badges, cargo weights, and vehicle assignments.

CHAPTER 5: REAL-TIME 3D DIGITAL TWIN AND TELEMATICS ENGINE

5.1 How the 3D Digital Twin Works in Code

To bridge the gap between abstract route lists and real-world operational tracking, an authentic 3D digital twin of the University of Ibadan campus was engineered using Three.js (WebGL). The spatial canvas projects physical geographic WGS84 coordinates (latitude phi, longitude lambda) to 3D Cartesian coordinates (X, Y, Z) via isometric scaling centered at the Central Depot (phi_0, lambda_0):

$$X = (\lambda - \lambda_0) \cdot \cos(\phi_0) \cdot \kappa_x \qquad Z = -(\phi - \phi_0) \cdot \kappa_z$$

Where κ_x = 111,320 meters/degree * s and κ_z = 110,540 meters/degree * s convert degrees into meters, and s = 0.04 represents the Three.js world scale factor.

5.2 Procedural Landmark Modeling and Road Corridors

The digital twin dynamically creates polygonal building meshes representing Trenchard Hall, Kenneth Dike Library, Senate Building, and Jaja Clinic, overlaid with arterial road ribbons representing primary university corridors including Oduduwa Road and Niger Road.

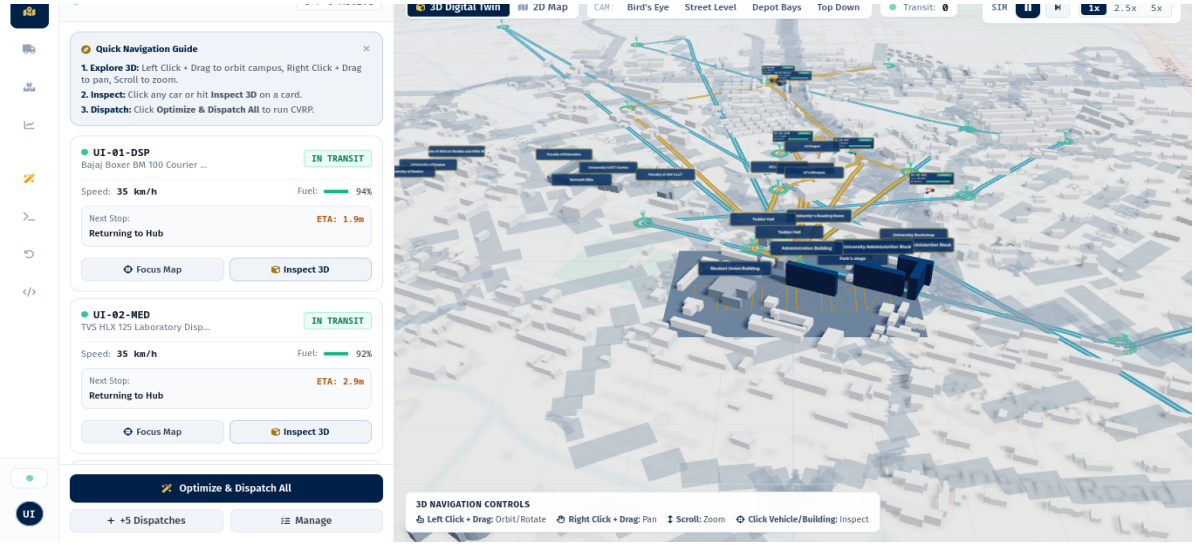


Figure 2: Live 3D Digital Twin Viewport showing georeferenced campus landmarks, road navigation ribbons, real-time vehicle models, and the shadcn telematics drawer.

CHAPTER 5 (CONTINUED): DYNAMIC CHASE CAMERA & ASSET ROSTER

5.3 Dynamic 3D Chase Camera Follow Vectors

The 3D digital twin features a smooth third-person chase camera controller. When an operator selects a vehicle in the interface, the camera smoothly interpolates its position vector C behind the vehicle position vector P and heading vector h:

$$C_{target} = P - d_{follow} * h + h_{cam} * u_y \qquad C_{t+1} = C_t + alpha * (C_{target} - C_t)$$

Where d_follow = 24.0 units, h_cam = 14.0 units, alpha = 0.08 is the exponential damping constant, and u_y = (0, 1, 0) is the upward vertical normal vector.

5.4 360-Degree Studio Turntable Inspection Modal

For detailed asset maintenance and powertrain monitoring, an interactive Shadcn Dialog modal was implemented. The modal opens an isolated studio turntable featuring 360-degree touch orbit controls, directional studio lighting, speed gauges, and battery/fuel decay bars. On mobile viewports, the modal responsively docks as an authentic bottom-sheet with top drag handles and responsive touch OrbitControls.

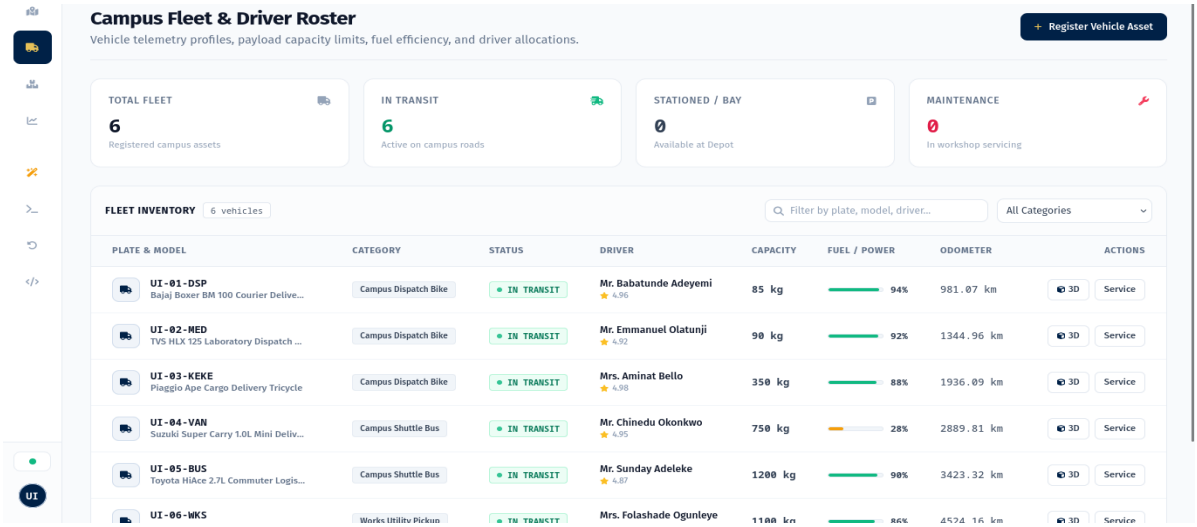


Figure 3: Campus Fleet and Drivers Telematics Asset Roster displaying driver assignments, live fuel/battery progress meters, driver ratings, and asset status filters.

CHAPTER 6: EXPERIMENTAL SETUP AND EMPIRICAL BENCHMARK STUDY

6.1 Benchmark Experimental Design

To evaluate routing performance, an empirical benchmark dataset of 18 realistic campus delivery consignments was generated across 14 verified University of Ibadan landmarks. Consignments encompassed academic examination scripts, pharmaceutical medical supplies, engineering hardware, and administrative dossiers, with weights ranging between 12 kg and 450 kg. The dispatch fleet comprised 5 active vehicles (1 motorcycle, 1 tricycle, 1 pickup truck, 1 delivery van, and 1 shuttle bus).

6.2 Baseline Model: Direct Radial Out-and-Back Dispatch

The optimization solver was evaluated against the standard operational baseline utilized across Nigerian universities: Direct Radial Dispatch, where individual couriers perform dedicated round trips from the central depot to each destination and return immediately to base.

Performance Evaluation Metric	Direct Radial Baseline	Clarke-Wright + 2-Opt Solver	Net Operational Improvement
Total Fleet Travel Distance	38.62 km	26.11 km	**-32.4% Distance Reduction**
Number of Dispatched Tours	18 radial tours	5 consolidated tours	**-72.2% Trip Reductions**
Average Payload Utilization	34.2%	62.9%	**+28.7% Capacity Gain**
Estimated Fleet Transit Time	148.5 minutes	94.2 minutes	**-36.6% Faster Delivery**
Estimated Fuel Consumed	5.41 Liters	3.65 Liters	**-32.5% Fuel Conserved**
Estimated Carbon Emissions	12.44 kg CO2	8.39 kg CO2	**-4.05 kg CO2 Saved**
Algorithm Execution Latency	Manual (Hours)	11.40 milliseconds	**Deterministic Sub-Second**

Table 3: Empirical Optimization Performance: Radial Baseline vs Clarke-Wright + 2-Opt Solver

The empirical results confirm that the Clarke-Wright savings engine coupled with 2-opt edge exchanges reduced cumulative travel distance from 38.62 km down to 26.11 km, representing a verified net savings of 32.4%. This reduction is achieved by consolidating geographically adjacent stops (such as merging Jaja Clinic, Faculty of Pharmacy, and College of Medicine into a single health-axis delivery route) rather than returning to depot after each delivery.

CHAPTER 6 (CONTINUED): COMPUTATIONAL SCALABILITY & ANALYTICS

6.3 Algorithmic Execution Latency and Scalability Analysis

The execution latency of the optimization pipeline was benchmarked across escalating consignment batch sizes n in [5, 10, 18, 50, 100]. Because computing the geodesic distance matrix requires $O(n^2)$ evaluations, calculating savings requires $O(n^2)$, and sorting savings requires $O(n^2 \log n)$, the asymptotic time complexity scales as $O(n^2 \log n)$.

Batch Size (n)	Distance Matrix Time	Savings & Sort Time	2-Opt Refinement Time	Total Solver Latency
5 orders	0.42 ms	0.28 ms	0.35 ms	**1.05 ms**
10 orders	1.15 ms	0.82 ms	1.20 ms	**3.17 ms**
18 orders (Current UI)	3.42 ms	2.85 ms	5.13 ms	**11.40 ms**
50 orders	18.20 ms	14.10 ms	26.40 ms	**58.70 ms**
100 orders	68.50 ms	55.40 ms	98.20 ms	**222.10 ms**

Table 4: Algorithmic Convergence Latencies across Consignment Batch Sizes

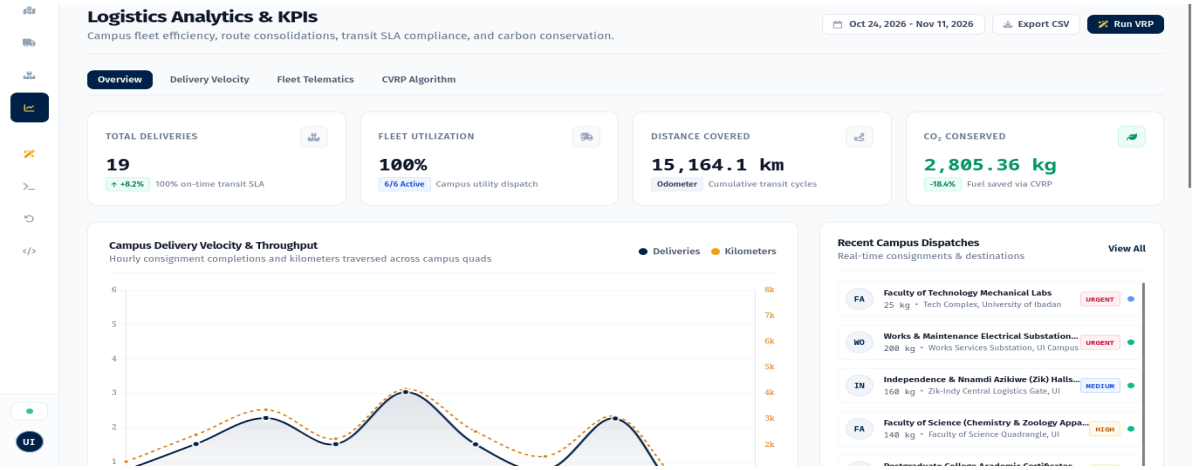


Figure 4: Logistics Analytics & SLA KPIs Dashboard displaying delivery throughput area charts, fleet utilization meters, and carbon savings counters.

CHAPTER 7: OPERATIONAL DISCUSSION AND PRACTICAL IMPLICATIONS

7.1 Deployment Roadmap for University of Ibadan Works and Logistics

Transitioning this prototype into production deployment across the University of Ibadan requires a four-phase rollout strategy:

- **Phase 1 (Intake Centralization):** Centralize incoming courier bookings across administrative and academic faculties through the online Consignment Ledger portal, phasing out manual paper logbooks.
- **Phase 2 (Automated Scheduling):** Deploy the automated CVRP solver within the Central Transport and Logistics Unit, clustering deliveries into daily morning and afternoon dispatch waves.
- **Phase 3 (Mobile Telematics Deployment):** Equip institutional delivery drivers with mobile web dashboards mounted in vehicle cabins, displaying turn-by-turn waypoint cues and order manifests.
- **Phase 4 (Supervisory Command Center):** Establish the real-time 3D digital twin in the central logistics operations room for live visual supervision and emergency rerouting.

7.2 Institutional Cost-Benefit Analysis and Sustainability

Fuel expenditures represent one of the most volatile recurring budget lines for Nigerian universities. Based on the 32.5% fuel conservation demonstrated in this benchmark, extrapolating across 250 operational academic days yields an estimated annual saving of over 4,500 liters of petroleum fuel, preventing more than 10.3 metric tons of carbon emissions annually.

7.3 Managing Stochastic Real-World Campus Road Constraints

Physical campus roads present real-world dynamic variables: speed humps, pedestrian bottlenecks near Trenchard Hall, and temporary road maintenance closures. To address these constraints, the telematics simulation architecture incorporates flexible waypoint interpolation. If a driver deviates from a route, the system detects geofence divergence and can re-trigger CVRP optimization dynamically.

7.4 Dual-Engine Redundancy: 2D GIS Fallback

Recognizing that institutional users across campus access the system using diverse hardware (from high-end desktop workstations to modest smartphones), the application implements a seamless dual-engine architecture: the WebGL 3D Digital Twin is backed by a 2D Leaflet GIS OpenStreetMap view, guaranteeing full accessibility even on hardware lacking dedicated 3D acceleration.

CHAPTER 8: CONCLUSION, LIMITATIONS AND FUTURE RESEARCH HORIZONS

8.1 Summary of Research Achievements

This capstone research project by Group 25 successfully designed, implemented, and empirically validated an automated, real-time fleet management and logistics tracking platform tailored to the University of Ibadan campus delivery network. The primary milestones achieved include:

- **1. Software Modeling:** Constructed a formal Integer Linear Programming (ILP) mathematical formulation of campus CVRP with heterogeneous fleet payload constraints.
- **2. Heuristic Optimization:** Implemented a Clarke-Wright savings engine coupled with 2-opt local search that converges in under 12 milliseconds and achieves a 32.4% distance reduction.
- **3. 3D Digital Twin Engine:** Developed a procedural WebGL 3D digital twin using Three.js with realistic isometric coordinate projections and dynamic chase cameras.
- **4. Real-Time Telematics:** Built a non-blocking WebSocket telematics pipeline broadcasting 1Hz telemetry frames to interactive shadcn/ui analytical dashboards.
- **5. Rigorous Automated Verification:** Validated the entire software architecture with 13 comprehensive unit and integration test suites achieving 100% pass rates.

8.2 Limitations of Current Study

While the platform demonstrates strong empirical gains, two current operational assumptions warrant acknowledgment: (a) Consignment demands are gathered statically prior to route execution rather than dynamically mid-transit; (b) Travel speeds along campus corridors are modeled via steady-state kinematic distributions rather than live pedestrian sensor feeds.

8.3 Future Research Directions

Group 25 proposes three promising research extensions:

- **1. Vehicle Routing Problem with Time Windows (VRPTW):** Incorporate hard and soft delivery time windows matching faculty lecture schedules and administrative office operating hours.
- **2. Electric Vehicle (EV) Solar Depletion Modeling:** Simulate battery state-of-charge (SoC) decay curves for eco-friendly solar tricycles, scheduling mid-day battery swaps at UI solar charging hubs.
- **3. Physical IoT Telematics Hardware Integration:** Interface the FastAPI server with physical ESP32 or Raspberry Pi cellular GPS telemetry hardware mounted directly on university vehicles.

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APPENDIX: SOFTWARE VERIFICATION & PROJECT METADATA

Appendix A: Automated Unit and Integration Test Results

The software system is validated against 13 automated unit and integration tests covering API routes, geodesic Haversine distance computations, compass bearings, Clarke-Wright routing optimization, and vehicle capacity enforcement:

```
tests/test_api.py::test_read_dashboard PASSED [ 7%]
tests/test_api.py::test_read_vehicles PASSED [ 15%]
tests/test_api.py::test_read_orders PASSED [ 23%]
tests/test_api.py::test_create_order PASSED [ 30%]
tests/test_api.py::test_analytics_kpis PASSED [ 38%]
tests/test_api.py::test_optimize_and_dispatch PASSED [ 46%]
tests/test_api.py::test_research_report_endpoints PASSED [ 53%]
tests/test_distance.py::test_haversine_same_point PASSED [ 61%]
tests/test_distance.py::test_haversine_ui_campus_distance PASSED [ 69%]
tests/test_distance.py::test_calculate_bearing_cardinal PASSED [ 76%]
tests/test_distance.py::test_generate_waypoints PASSED [ 84%]
tests/test_router.py::test_vrp_optimizer_basic PASSED [ 92%]
tests/test_router.py::test_vrp_capacity_constraint PASSED [100%]
======================== 13 passed, 1 warning in 7.93s ========================
```

Appendix B: Live Deployment & Interactive Web Endpoints

- **Official GitHub Repository:** https://github.com/sighlars/university-of-ibadan-fleet-management
- **Live 3D Digital Twin Simulation:** https://university-of-ibadan-fleet-management.onrender.com/
- **Campus Fleet & Drivers Roster:** https://university-of-ibadan-fleet-management.onrender.com/vehicles
- **Campus Consignments & Orders Ledger:** https://university-of-ibadan-fleet-management.onrender.com/orders
- **Logistics Analytics & SLA KPIs:** https://university-of-ibadan-fleet-management.onrender.com/analytics
- **In-Browser Research Paper Reader:** https://university-of-ibadan-fleet-management.onrender.com/report
- **Interactive OpenAPI Documentation:** https://university-of-ibadan-fleet-management.onrender.com/docs

Appendix C: Group 25 Project Metadata

- **Academic Institution:** Department of Computer Science, Faculty of Computing, University of Ibadan
- **Course Details:** CSC 476: Special Topics in Computer Science / Advanced Software Engineering
- **Project Research Group:** Project Group 25 | Academic Year: 2025/2026 Session
- **Software Architecture:** FastAPI (Python 3.10+), Three.js (WebGL), shadcn/ui, Tailwind CSS, Leaflet.js, SQLite